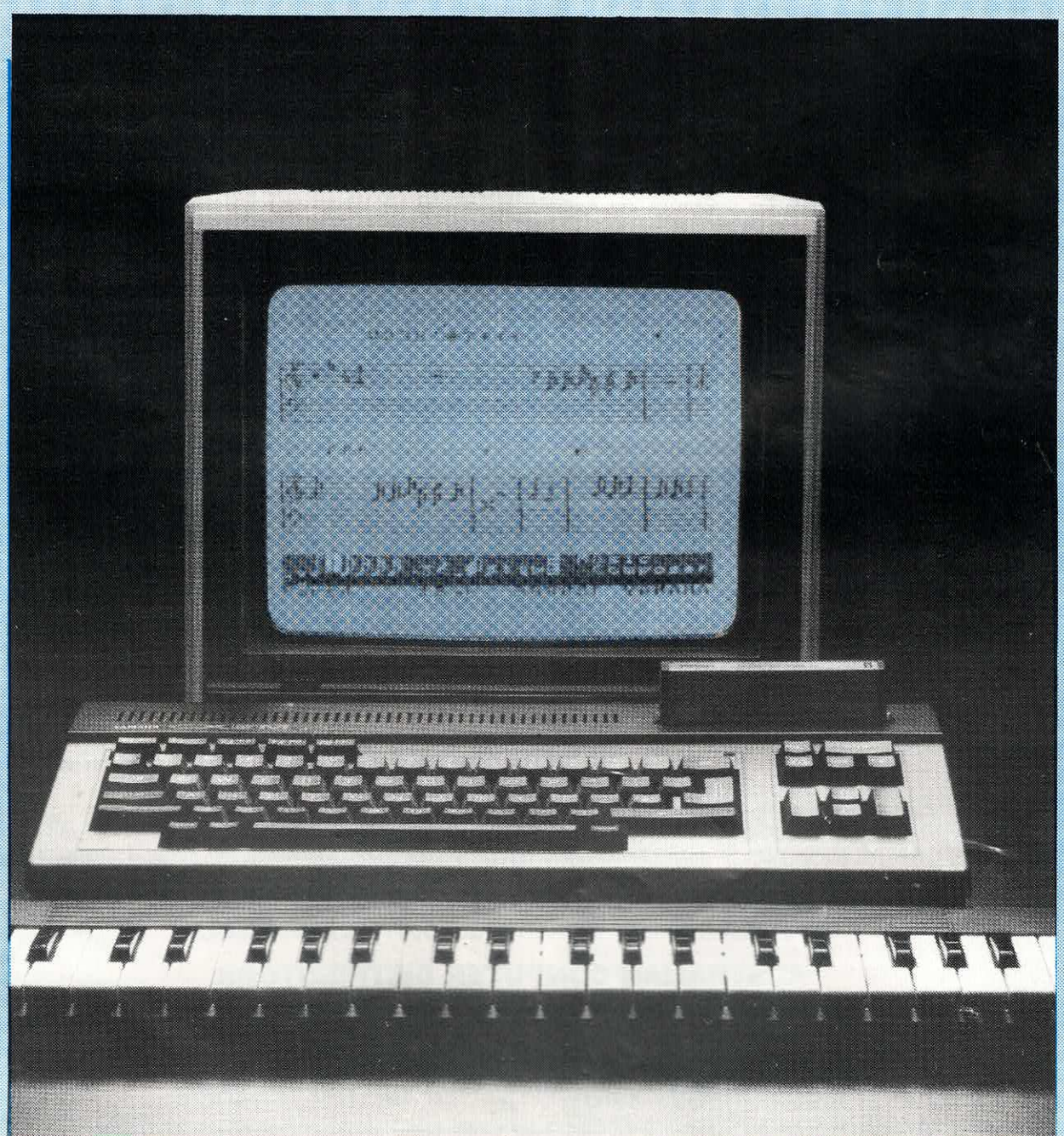
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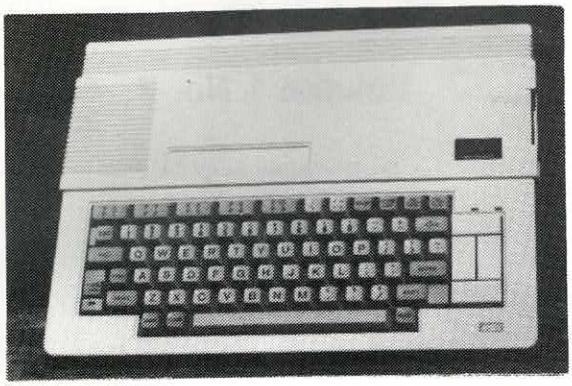
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MSX and SPECTRAVIDEO
COMPUTER FORUM
Magazine

Volume 1 No. 7

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EDITORIAL

The computer industry in this country is in a mess. Commodore recently canned 33 staff in Australia. Commodore Magazine is no longer. The C64 has been re-released as the 64C and the only real difference would seem to be a change of colour and moving the C to the other side of 64.

The new Atari computers are having a battle with the Amiga. Personally I think they are both overpriced. IBM clones are pouring into this country — some good, some bad. Pricing of computers is crazy, due to the weakness of the Australian dollar. Remember — even Australian built computers use imported components.

Some computer manufacturers have had their fingers burnt with home computers. Many dealers have folded their tents or moved into business computers. A number of users have shelved their computers due to lack of support and software.

The computer industry needs rationalising. A few months ago I would have predicted three standards emerging — IBM compatibles, MacIntosh/Amiga compatibles and MSX. The way things are going, you can probably discount all but the first one.

Happy computing?

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CX5M — COMPUTER OR MUSICAL INSTRUMENT?



For the past few days we have been playing around with a Yamaha CX5M. With so many stories going around about this machine, we wanted to see for ourselves just what it was like to use as a computer. Elsewhere in this issue is a review of its music capabilities — but what about computing power?

The Yamaha CX5M is a 32K MSX computer with inbuilt FM Music Synthesizer. Our first impression was of a heavy machine, weighing nearly three kilos. This is due in part to the Synthesizer and also to the fact that there is a lot of metal in the computer. The whole thing is shielded inside with heavy gauge sheet metal for RF shielding and the keyboard is steel backed. This tends to give the keyboard a very solid feel with no bounce or flexing under heavy typing. The power supply is a separate unit rated at 240V and 30 Watts.

Looking around the computer, it is a standard MSX computer — apart from the connections to the Synthesizer on the left. There is one cartridge slot in the top and two joystick ports on the right-hand side. Across the back are the expansion, printer and cassette ports together with audio, video, TV and power sockets.

The top cartridge slot is MSX standard being a female 50 way card edge connector. The expansion port on the back, however, is a 50 way male PCB card edge. No chance of plugging any cartridges in here. To use this port for some other manufacturers' peripherals would require the optional CA-01 Single Cartridge Adaptor. This adaptor is also required if you wish to use the Yamaha FD-05 Disk Drive with the CX5M.

In the cartridge port is a reset button (microswitch) which is activated by inserting a cartridge or using your finger for manual reset.

MEMORY CAPACITY

The biggest criticism of the CX5M is memory capacity. This is made up from 4 only 8444 (MB81416-12) RAM chips which gives 32K from &H8000 to &HFFFF. Some software will not work without the addition of a memory expansion cartridge.

We loaded Ghostbusters, which is a large program, without any hassle. Trying to load The Hobbit resulted in the computer 'freezing' and Barnstormer, which is marked as requiring 48K came up with "Not enough RAM: 48K required".

The Bank Street Writer cartridge gave no problems when using it with a data cassette. Some users have reported that this cartridge will not work with a disk system. This could be due to the fact that the Yamaha Disk system is double sided 720K format and we think that the area allocated to the File Allocation Table reduces the usable RAM just that much more.

DOCUMENTATION

The thing that really impressed me about the CX5M was the documentation. The Owner's Manual is 50 pages and covers the usual setting up procedures, loading and saving programs, inserting cartridges etc. Using the sound synthesizer is also covered.

The Reference Manual has 355 pages. Each Basic command has at least one page devoted to it. A tutorial section on programming techniques. Appendices contain maths functions, error codes, memory map, MSXDOS commands, MSX Disk Basic commands, I/O map and pinouts, some BIOS routines and RAM variables listing.

It would be nice to see this book being available separately — it is that good.

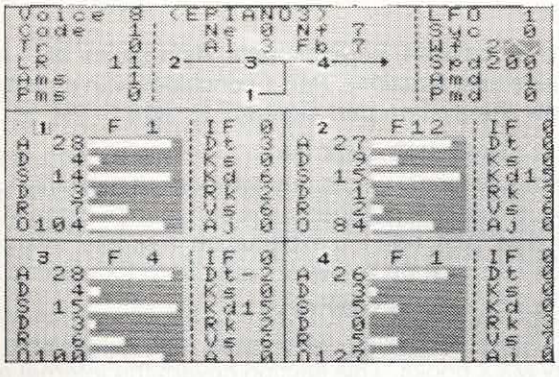
Erik Kowarski provides this review of the CX5M Music Computer from Yamaha.

The Yamaha basic system comprises — CX5M MSX computer with 32K RAM, 5FG-01 music synthesiser (fitted in special side slot inside the computer), YRM-102 FM Voicing Program, YRM-101 FM Music Composer, YRM-104 FM Music Macro. The last three items being software cartridges. Although the FM Music Composer can be used as it is, for most applications a Music Keyboard (YK-01 or YK-10) is also required. Recommended prices — CX5M computer with synthesiser and three software cartridges $1150, YK-01 $160, YK-10 $295.

At first glance, the price tag seems prohibitive, but this machine is no toy — the synthesiser alone is a slightly cut-down version of the famous DX-7, a synthesiser suitable for the most serious of musicians (retailing for well over $2000). Up to eight voices, each totally independent, can be generated at the same time. The CX5M is also equipped with MIDI (Musical Instrument Digital Interface), making it possible to control a wide range of external units such as other synthesisers, drum units, delay lines, etc.

The 5FG-01 can be used without supporting software, giving a kind of "super-Casio" layout, complete with auto bass/chord/rhythm (if you want it), or it can be played as a splittable-keyboard polyphonic synthesiser, but its main features lie in the three cartridges provided.

FM SYNTHESISER



The FM Voicing Program is, in effect, the synthesiser's control panel. Yamaha has chosen FM (Frequency Modulation) for its synthesisers, a system which combines four oscillators (in the 5FG-01) called operators in a variety of ways (algorithms) to generate complex waveforms unheard-of in the more conventional oscillator-filter style synthesisers. Sound creation is a little less predictable, but a few days of experimentation, aided by the thorough documentation provided in the YRM-102 manual, is bound to yield some very rewarding results.

Some features of the voicing program include a variety of controls related to dynamics and keyboard response, LFO controls (for vibrato, tremolo etc.), individual five-stage envelope generators for each oscillator, and more. This instrument's strength lies in its ability to produce complex, "natural-sounding" voices with a very wide dynamic range. A noise generator is also included, but lacks the fine control of the rest of the unit, making some of the old "analogue-style" sounds harder to find than on conventional synthesisers. Forty eight user-programmed voices can be saved to tape or data cartridge, and another forty six factory presets reside in ROM. Any two of these can be accessed at any time, using the keyboard split facility, with seven-note polyphony on one side of the variable split point, and single notes on the other.



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The FM Music Composer is an 8-voice sequencer with all the facilities of a written score — repeats, dynamics, key and tempo changes, staccato, program (voicing) changes as well as notes and rests can be entered. There is also a transpose function. Entry is either from the computer or the Music Keyboard. The basic system has room for about 8000 notes, though loading in a set of forty eight user-programmed voices reduces this to about 7000. Notation is as close as possible to musical notation, though sadly lacking is the ability to join tails of short notes into one or two beat groups, as is the proper standard. Also "portamento" (glissando — continuously sliding pitch) cannot be written in, although it is available when playing manually.

MUSIC MACRO

The FM Music Macro is similar to MSX PLAY and SOUND commands allowing control of the 5FG-01 in BASIC, but is far more complex, to cater for all the features of the synthesiser. As a musician, I find this kind of program rather cumbersome, but it does open up the way for superior games, and music/video productions.

The whole package comes with thorough documentation in the form of the best BASIC handbook of them all, and four separate manuals for the synthesiser, voicing program, music composer and music macro. They are well written and easy to follow.

IMPRESSIONS

My impressions of this unit (and I've had mine for over a year) are pretty favourable, on the whole. The limited noise control is a slight handicap, and I miss the portamento on the music composer, but the degree of voicing and dynamic control, plus eight-voice polyphony, more than makes up for this. You should plug the (stereo) outputs into a decent hi-fi to get the full effect, and then you'll hear that rich, clear, "natural" sound which is the hallmark of FM synthesis: even the factory presets are mostly pretty good. Also, further expansion with MIDI enhances its appeal.

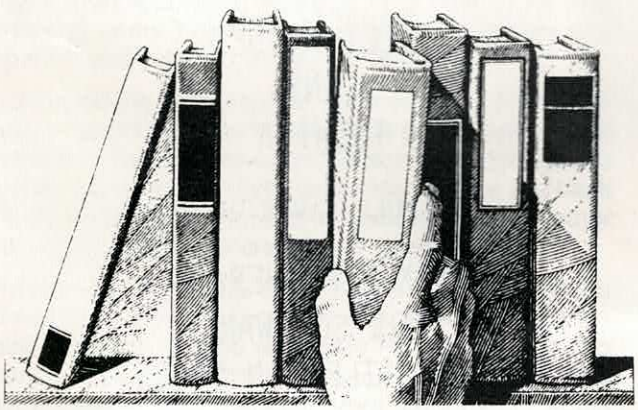
The main drawback is in the computer itself. With only 32K RAM, expansion to MSX-DOS or CP/M is a long way (and many dollars) off. This is certainly disappointing for such an expensive machine. It is also worth noting that the 5FG-01 and associated software is NOT disk compatible, and adding a disk drive will only be of use in non-music applications (though there is a way around this for saving voice data).

However, as a tape-based system or with disk BASIC, it is no different from the other MSX-1 computers. The sculptured keyboard feels nicer than most, and it comes with an external heavy-duty power supply with good ventilation for those hot summer months.

In summary, if you are interested in music and computers, this unit combines both in an economical and viable way. For me, as a professional musician and arranger, the benefits far outweigh the disadvantages, and the non-musical uses are a complete bonus. For the computer buff, it could be the easy way into the exciting world of music.

Yamaha is distributed in Australia by Rose Music.

BOOKS • BOOKS • BOOKS • BOOKS



THE MSX BOOK

by Paul Hoffman

This book is divided into three parts and appendices.

Part 1 YOUR MSX COMPUTER — This section discusses in plain language what is inside your computer, and the points which make the MSX standard. Also covered are the different types of software available (word processors, spreadsheets, data bases etc.) and explains their functions. Part 1 concludes with an outline of how the system can expand.

Part 2 LEARNING MSX BASIC — This section is the reason a beginner would buy this book. It explains, with copious example programs, the commands of the MSX language. Although the explanation is simple, even old hands can pick up a few wrinkles from the examples.

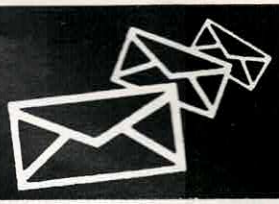
Part 3 LEARNING MSX-DOX — This covers the MSX Disk Operating System which is an area not covered by many MSX books. This section covers the internal and external commands of MSX-DOX and how to use them.

Appendices — MSX Basic command summary, MSX-DOS command summary, ASCII table, memory maps and I/O table and keyboard layouts.

This book can be recommended on the basis of scope rather than detail. The detail is sufficient to give a usable knowledge of an area of interest. Review copy supplied by McGraw-Hill Australia.

Publisher: Osborne McGraw-Hill
292 pages
Price $33.95

MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

CP/M ON HITBIT DISKS

Could you please let me know how I could run CP/M on my Spectravideo MSX 728 with a Sony HitBit disk drive as I have plenty of CP/M software available to me but at present cannot run it.

Daryl Nicklen
Craigieburn, Victoria

Many people have asked if CP/M can run on their MSX machines. Only Spectravideo supply CP/M with their disk drives. these work fine with other MSX computers but not other MSX drives. If any reader has managed to get CP/M running on drives other than SV — please let us know.

IT'S IN THE STARS

We were wondering is there anyone out there in Spectravideoland who has developed an astrological program. If there is, could you let us know what house system it uses. We have a SV328 D/D Disk Drive unit and would appreciate any help we could get. I heard that a disk has been produced in Australia that converts your Spectravideo to MSX, though it only works 'some of the time' — any comments?

P.J. Zalewski
Wellington, New Zealand

Any astrological types out there? The MSX/SV conversion program does, as you say, work some of the time. The version we have seen can only be used to convert Basic programs, as machine code programs need all of the output commands altered.

WORDPROCESSOR HELP

I have just received copies of your magazine and as a 328 owner find them a great answer to the frustrations caused by the lack of information from which I had been suffering — keep up the good work! I look forward to the promised machine language tutorials, and the DOS, in later issues.

Like Greg Weston (issue 2) I found the cassette file storage involved with Just Write Jnr cumbersome, and so tried VDO, a disc based CP/M public domain word-processor. This looks pretty good, but requires an 80 column card, which I lack. I am trying to modify it for 40 column but it is proving a tough assignment as I only have the .COM file (any help in this matter would be gratefully received). Without sufficient cash for Wordstar I settled for Spectratext which costs about $NZ50 from Piaggi Computer Systems, 305 Mary Street, Thames, N.Z. It is written in Basic and so is not superfast, but

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intelligent use renders its speed satisfactory for writing letters etc. It comes on disk or tape, and handles both disk and tape storage, 40 or 80 column, any line length. It lacks the scroll facilities of Just Write Jnr, but enables printer codes, and comes with an easy-to-understand and succinct operating handbook of 10 pages. An SV328 or 318 with 16K expansion card is required. I hope this information is of use. There is a chess program under CP/M in the public domain.

Jim Megeniz
Coromandel, New Zealand

Thanks for the information, Jim.

DISK AND OTHER QUERIES

I am using a SV328 with two disk drives and 80 column card and would like to know how to make the computer boot up to the 80 column card. Another problem is that SVFRMT will not operate while using the 80 column card. Both of these would need an alteration to the programs on the disk. Also COMMON appears to be a reserved word. How is this used? CHAIN is listed as a reserved word but does not appear to be so as the computer accepts it as a variable name. Finally, is there an assembler available to run under disk basic?

R.J. Thomson
Nambour, Queensland

The 80 column boot was covered in our first issue, and involves adding a line containing WIDTH 80 to the IPL program. There does appear to be a clash between SVFRMT and the 80 column card with only the first two tracks formatting. Has anybody fixed this? Much of the SV manuals were borrowed from existing text covering MBASIC and BASICA and contain some commands which are not supported in SV Basic — such as CHAIN and COMMON. We are not aware of a commercial disk based assembler for the SV. You might get some joy by contacting the user groups listed in Computer Forum.

PEEKING THE KEYBOARD

After reading your reply to S. Dimitrio on the SELECT and PRINT keys, I tried PEEKing &HFD83 and got nowhere. After further searching the memory locations around &HFD83, I finally located the SELECT and PRINT key results in &HFD88.

By this time, my curiosity was well and truly up. I continued looking for all the keys and their return codes. The end result is the following program which shows which memory locations are used, the hexadecimal returns and a binary representation which should make it clear to all and sundry why those hex codes are returned. For the MSX computers, substitute &HFD80 in lines 20 and 80 with &HFBEO.

```
10 FOR I=0 TO 10                    :'there's 11 locations
20 A=PEEK(&HFD80+I)                 :'look at memory
30 IF A<>255 THEN GOSUB 60          :'jump if key pressed
40 NEXT : GOTO 10                   :'all viewed, go again
50 '-----------------------------------------------
60 A$=BIN$(A)                       :'convert to binary
70 A$=A$+STRING$(8-LEN(A$),"0")     :'fill in zeros
80 PRINT HEX$(&HFD80+I);"   ";      :'print mem location
90 PRINT HEX$(A);"   ";             :'print hex data
100 PRINT A$ : RETURN               :'print bin and go back
```

Happy hacking

Graham Crawford
Reynella, South Australia

This is the type of letter we love to receive. Graham has read something in the magazine which fired his enthusiasm and he is sharing the results with other readers. Thanks Graham.

READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

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MAILBOX

GRAPHICS PRINTER HELP

Congratulations on a fine magazine — I am glad now that I took the plunge on a twelve month subscription. I have a SV328 with cassette and have recently purchased a SV901 Printer and wordprocessing software. Printing text is just fine, but graphics is a problem.

I copied the SCREEN DUMP routine by Rod Johnson in Volume 1 No. 4, but cannot work out the line changes for the SV901. Can anyone help me on this? How is the screen dump loaded in relation to a graphics program? Being a raw beginner to computers, I am not familiar with these procedures. Hoping that you or one of your readers can help me out.

Ted Kobelt
Glen Waverley, Victoria

We were not able to test the Screen Dump program on a SV901 printer. A long shot would be to look at the operating manual for a Tandy Line Printer VII or Dick Smith GP80. The style of print is very similar to the SV901 and possibly the graphics are the same. The Tandy manual gives a good explanation of how the graphics work. Can anyone help Ted further with this problem?

AND AGAIN

Allow me to congratulate you on your magazine. COMPUTER FORUM is the first, most practical mag I've seen for SV users.

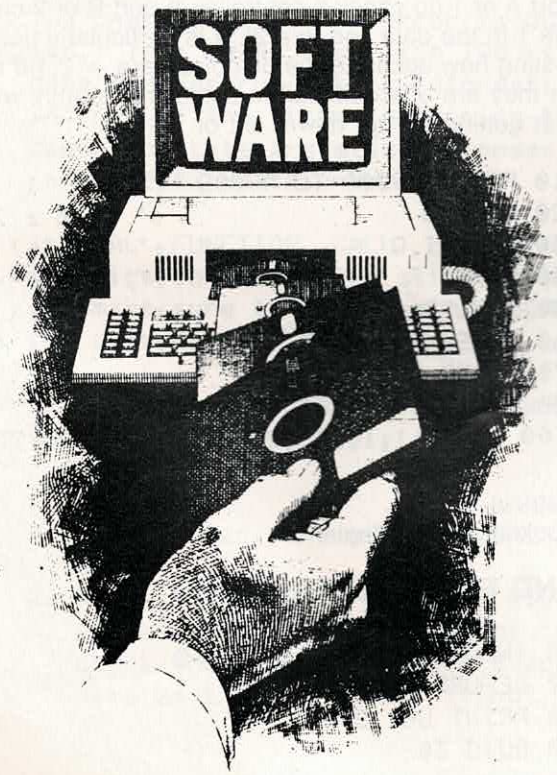
Could you inform me as to how I can employ Rod Johnson's screen dump program to persuade my 328's Graphics Tablet to dump to my Epson LX80. Right at the moment I get about three characters that closely resemble Chinese arithmetic.

Is the Panda Soft and FDS Software (Software News & Reviews CF No. 4) suitable for the SV328?

Stuart Dall
Essendon, Victoria

The software you mention is suitable for SV328 except for Mt Base and Aerobat. The software available with the SV Graphics Tablet was apparently available in two versions — one for the SV901 and one for other prints. We can only suggest that you read the graphics section of your printer manual and, if possible, try your system with an SV901 to see which version of software you have.

Software NEWS & REVIEWS



BATTLESHIP CLAPTON II (MSX Tape) by Toshiba

(Picture of screen not available for publication)

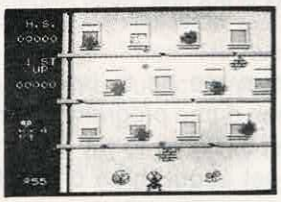
This is an alien invader-type game with some differences. As usual, you control the defending space fighter except that, apart from moving sideways you can move forward to avoid the missiles which the aliens can fire sideways at you as they go past. Of course, staying near the top of the screen gives you less chance to avoid the aliens and their head on missiles. After a certain score is achieved, a mother ship appears on screen which can be destroyed for bonus points.

Written in machine code, this game has good graphics and smooth movement and is good value for money.

Review copy by Peter J. Fox Computers
Price: $15.00

Software NEWS & REVIEWS

FRANTIC FREDDY (SV Cartridge) by Spectravideo



This cartridge is a one or two-player game where you control Freddy who must put out fires, avoid falling debris and save the cats.

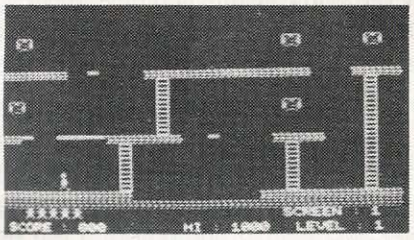
Shooting the falling fire particles with water bursts gains 10 points. Each stage of the game has a countdown starting from 1000. When you complete each level, this figure is taken into account and you are given points — the quicker you complete the levels the more points you get.

On the second screen, the magenta fires, which moves toward Freddy, cannot be put out until the screen is cleared of red fires. On the higher levels, you can enter doors to get to different floors.

Review copy by Peter J. Fox Computers
Price: $49.00

GAME PACK 7 (MSX & SV Cassette) by FDS Software

SVPAK7 and MSXPAK7 consist of two programs — BOMB SCARE and DUNGEON ADVENTURE.



BOMB SCARE is a platform game where you must climb to the platforms and defuse the bombs before you can get to the next screen. The ladders are easy enough to climb, but the moving platforms and conveyor belts are a little tricky to negotiate. There are six screens to this game which is very addictive.

DUNGEON ADVENTURE is a classic adventure program where you give instructions to the computer about which way to go in the dungeon. You will be confronted by monsters and spiders and, if you are lucky, find some treasures in your quest for the Great Orb.

Price: $16.00.

HINTS & TIPS

KEYBOARD TOGGLE (MSX)

The following is similar to a previous routine for turning the caps lamp on/off, but switches the keyboard between upper and lower case letters at the same time —

```
10 POKE 64683!,255 : OUT &HAA,0
20 FOR I=1 TO 200 : NEXT
30 POKE 64683!,0 : OUT &HAA,255
40 FOR I=1 TO 200 : NEXT
50 GOTO 10
```

Andrew Brown
Rosanna

TESTING JOYSTICKS (MSX)

This program is useful for testing joysticks. It continuously displays the value of the position of the controller in port A or 1 (to check a controller in port B or 2 change the 1 in the data line to a 2). It is particularly useful in testing how accurate the diagonals are — 2,4,6 and 8 as they are often difficult to achieve smoothly without first getting an up, down, left or right.

```
10 FOR I=49000 TO 49008  :'      1
20 READ Q                :'    8 : 2
30 POKE I,Q              :'     \:/
40 NEXT                  :'   7--0--3
50 DEFUSR=49000          :'     /:\
60 A=USR(0)              :'    6 : 4
70 PRINT PEEK(50000)     :'      5
80 GOTO 60
100 DATA 61,1,205,213,0,50,80,195,201
```

Patricia Perris
Auckland, New Zealand

AND FOR SPECTRAVIDEO

```
10 INPUT "STICK NUMBER";S
20 DEFUSR=&H3206
30 PRINT USR(S)
40 GOTO 30
```

COMPUTER FORUM
Magazine
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BASIC TUTORIAL — Part 7

ENTERING PROGRAMS INTO YOUR COMPUTER

When you are typing a program into your computer (for example, the listings in this magazine), there are certain things which can make the difference between the program running first time and hours of frustrating checking.

The first of these are spelling mistakes. It does not matter if you make mistakes typing anything which is within quotation marks except when they are part of a PLAY or DRAW command. If you mispell a command, then a SYNTAX ERROR will result. The command can be entered in upper or lower case and the computer will automatically convert these to upper case.

Another common error is not pressing ENTER before typing the next line number. This can happen when one line ends at the right hand side of the screen and the cursor has moved down to the next line. Looking at it on screen will give the appearance that all is well, because the ENTER key does not display on the screen.

The next most common error is OUT OF DATA. For example OUT OF DATA IN 100. This means that line 100 is attempting to read a certain amount of data and has run out of information. The error is not in line 100, but in one of the lines containing the DATA which is being read. Typical causes of this are missing commas or incorrectly entered quotation marks in the data statements.

ILLEGAL FUNCTION CALL can be caused by the computer encountering a graphics command while still in text screen mode. Check the program flow and make sure that a graphics SCREEN command is in a line which precedes the offending command.

There are many other problems which can occur in typing programs, but the general rules are -

1. Use a ruler as a guide to the line being typed.

2. Where possible, use the WIDTH command to make the text on screen appear the same as that of the printout. This will enable you to check that the same letters appear under each other on the screen the same as the printout. If not, something has been missed or added in.

3. Check each line as it is entered. This is far easier than scrolling through the program after everything is finished.

4. Watch out for commas and full stops. These are the most difficult to reproduce in a printed medium. The general rule is that it can only be a fullstop if contained in a string within quotes or used as a decimal point in a numeric value.

5. Make sure that every left hand bracket has a matching right hand partner and the same with quotation marks.

PROGRAMMING TECHNIQUES

The most important factor in writing your own programs is PROGRAM FLOW. This does not mean that it must be easily read by you, but by the computer. To understand this, it is necessary to examine how a computer reads a program.

Because of GOTO, GOSUB, READ and FOR/NEXT commands, the computer does not necessarily start at the lowest line number and proceed line by line until the end of the program. For example when we read the command GOSUB 1000 in line 250, we draw on our memory and experience to know that line 1000 is greater than 250 and start from there to find line 1000. The computer, on the other hand, starts looking for line 1000 by going back to the first line and checks through all of the line numbers until 1000 is found. We can make this task easier by placing the subroutines, particularly the time-critical ones, at the start of the program.

Another point which will slow a program down is to have the computer check information unnecessarily. For example, we would use a series of IF/THEN commands to determine if a number is in a certain range. In this case check for the 'most likely result' first. The following example will illustrate these two points.

THIS could be re-written as THIS

```
100 INPUT "ENTER NUMBER (1-9)";N
110 REM WANTED NUMBER IS 5
120 IF N<5 THEN GOSUB 320
130 IF N>5 THEN GOSUB 340
140 IF N=5 THEN GOSUB 300
150 GOTO 100
300 PRINT "CORRECT"
310 RETURN
320 PRINT "TOO LOW"
330 RETURN
340 PRINT "TOO HIGH"
350 RETURN
```

```
10 GOTO 300
100 PRINT "CORRECT"
110 RETURN
120 PRINT "TOO LOW"
130 RETURN
140 PRINT "TOO HIGH"
150 RETURN
300 INPUT "ENTER NUMBER (1-9)";
310 REM WANTED NUMBER IS 5
320 IF N=5 THEN GOSUB 100 : END
330 IF N<5 THEN GOSUB 120 : GOTO 300
340 IF N>5 THEN GOSUB 140 : GOTO 300
```

The speed difference between these two programs is not evident, but there would be a marked change in speed between larger programs written the same way.

The following commands can be used to assist in writing and entering programs into the computer.

The AUTO command

This is the AUTOmatic line numbering command. It overcomes the need to enter the line number when typing in a program. It takes the following forms -

AUTO	Start line numbers at 10 with an increment of 10
AUTO 100	Start line numbers at 100 with an increment of 10
AUTO 100,20	Start line numbers at 100 with an increment of 20

If AUTO creates a line number that already exists (such as when adding lines to the start of a program), an asterisk * will be displayed next to the line number. If you do not wish that line to be overwritten, then press the ENTER key to go to the next line.

AUTO can be cancelled by holding down the CTRL key and pressing STOP. The line number just generated will not be saved.

The DELETE command

This command is used to DELETE a line or block of lines as specified. It takes the forms -

DELETE 10	Delete only line 10
DELETE 10-30	Delete lines 10 to 30 inclusive
DELETE -30	Delete from the start of the program up to and including line 30
DELETE 30-	Delete line 30 and all lines higher than 30
DELETE.	(DELETE followed by a full stop) delete the last line entered

The first example is not used because typing the line number and pressing ENTER will have the same effect.

The RENUM command

This command is used to RENUMber the lines of a program.

It takes the form RENUM a,b,c in which the program will be renumbered using the new line number (a) starting with the old line number (b) and use an increment of (c).

RENUM	Renumber the existing program using the new first line number 10 and an increment of 10
RENUM 200	Renumber the existing program using the new first line number 200 and an increment of 10

```
RENUM 100,,5        Renumber the existing program using the new first
                    line number 100 and an increment of 5
RENUM 210,120,5     Renumber that part of the existing program commen-
                    cing with line 120 using the new first line number
                    210 and an increment of 5
```

```
example (old program)     100 REM THIS IS A TEST
                          150 REM
                          170 REM
```

the command RENUM 200,100,10 is used and the program now reads

```
                          200 REM THIS IS A TEST
                          210 REM
                          220 REM
```

The KEY command

On power-up the function keys F1 to F10 are given pre-determined commands. This command allows you to change those functions. This can be useful if you have a great number of repeats of the same command. For example, if a program contains lines of DATA statements changing F1 to DATA would save some typing. To do this type

KEY1,"DATA" and press ENTER

Using AUTO and function key F1 now makes it a lot easier to enter the lines of data. The procedure being that the computer will display a line number, you press F1 and continue to enter the values for the data statements.

The TRON and TROFF commands

These are the TRACE ON and TRACE OFF commands and can be used to follow the program flow as it is executing. Each line number is displayed in square brackets as it is being executed. This can be useful when the computer appears to get stuck at a certain point in the program or doesn't seem to do what the program requires.

The procedure is to type TRON before you RUN the program. Type in this program

```
100 PRINT "LINE 100"
200 PRINT "LINE 200"
300 PRINT "LINE 300"
```

Now type TRON and press ENTER. Type RUN and press ENTER and the screen

should look like this

```
[100]LINE 100
[200]LINE 200
[300]LINE 300
```

DO NOT FORGET to type TROFF and press ENTER to cancel the TRACE ON function before running another program.

TRON and TROFF should only be used in text mode, as the results are unpredictable on a graphics screen.

MANAGING VARIABLES

Several commands exist to allow for the re-organisation and management of variables used in programs. The first of these allows you to change from alphabetical to numeric and back.

The ASC command

Each printable character used by the computer has a value according to the ASCII Table. This table, the American Standard for Computer Information interchange, is one of the very few standards used by all home computers.

The manual which accompanied your computer should contain this table. In general, the numerals 0 - 9 are ASCII codes 48 to 57, the upper case alphabet is covered by 65 to 90 and the lower case alphabet has codes 97 to 122.

When a character is stored in memory, it is actually the ASCII value which is held there. For example, if the character A is stored in a certain memory location, PEEKing that location will reveal 65 which is the ASCII value of the capital letter A.

The ASC command is used to convert alphabetical variables to their numeric equivalent according to the ASCII table. It takes the form ASC(X$) where X$ is the character for which the ASCII value is required.

Try this program

```
10 INPUT "ENTER A CHARACTER";I$
20 PRINT ASC(I$)
30 GOTO 10
```

This program will print the ASCII value of the character entered.

The CHR$ command

This command is the reverse of the ASC command. It takes the form CHR$(X) where X is the ASCII value for which you require the corresponding letter. Try this program

```
10 INPUT "ENTER A NUMBER FORM 65 TO 90";I
20 PRINT CHR$(I)
30 GOTO 10
```

The STR$ command

This command is used to convert a numeric variable to a string variable. It takes the form STR$(X) where X is the numeric variable or number to be converted. For example -

```
10 X=10
20 A$="ABC"
30 A$=A$+STR$(X)
40 PRINT A$
```

Displays ABC 10

This would have uses in such things as checking for correct input of a date. For example -

```
10 INPUT "ENTER DATE (DDMMYY)";D
20 IF LEN(STR$(D)) <> 7 THEN 10
```

Notice in both of these examples that the length of the converted variable is one greater than the actual length of the number. This is due to the fact that the sign (+ or -) is taken into account. Whenever a numeric variable is displayed, allowance is made for the sign which is only printed if it is minus.

The VAL command

This is the reverse of the STR$ command and is used to convert a string variable containing numbers to a numeric value. It takes the form VAL(X$) where X$ is a string variable or alphabetical characters in quotes. It will return the value of any numbers in the string which are not preceded by alphabetical characters. For example -

```
10 A$="ABC1"
20 B$="1ABC2"
30 C$="123ABC"
40 PRINT VAL(A$)
50 PRINT VAL(B$)
60 PRINT VAL(C$)
```

Type RUN and the program should display -

```
 0
 1
 123
```

There are other commands which allow alteration of variables.

The SWAP command

This command can be used to SWAP the value or meaning of variables and can be used with numeric or alphanumeric variables. It takes the form SWAP variable1,variable2 where variable1 and variable2 are numeric or string variables separated by a comma. For example

```
10 A=10                    10 A$="ABC"
20 B=20                    20 B$="DEF"
30 SWAP A,B                30 SWAP A$,B$
40 PRINT A                 40 PRINT A$
50 PRINT B                 50 PRINT B$
```

When these are run, the computer will display

```
 20                        DEF
 10                        ABC
```

The INSTR command

This command can be used to examine a part of a string variable to see if it contains certain characters or combination of characters. The value returned is the location within the string where the character or combination first appears. It takes the form INSTR(A$,B$) where A$ is the one being examined to see if it contains B$. For example -

```
100 A$="ABCDEFG"
110 PRINT INSTR(A$,"CDE")
120 PRINT INSTR(A$,"MNO")
130 PRINT INSTR(A$,"ABCDEFGHIJKL")
```

Line 110 will display 3 because CDE starts at position 3 in A$
Line 120 will display 0 because MNO is not contained in A$
Line 130 will display 0 because ABCDEFGHIJKL is longer than A$

INSTR also allows you to specify where in the string to start looking. For example -

```
100 A$="ABCDABCDABCD"
110 PRINT INSTR(6,A$,"ABC")
```

will display 9 because we have specified to start at the 6th character of A$ and look for ABC which starts at position 9.

The FRE ('' '') OR FRE (A$) command

This command allows us to check how much string space is still available. The string space is allocated using the CLEAR command and is a block of memory set aside for use by string variables. If a CLEAR command is not used, then the computer sets aside room for 200 characters.

This space is not used by variables which are given a value within the program, such as A$="ABC" but is used for variables which are to be inserted. For example -

```
10 CLEAR 200
20 A$="ABC"
30 PRINT FRE("")
40 INPUT B$
50 PRINT FRE("")
```

When this is run, notice how line 30 still displays 200 but line 50 will display 200 minus the length of the variable input at line 40.

MORE SOUND COMMANDS

The PLAY () command

Because the Programmable Sound Generator is actually a separate CPU, it is possible for the main CPU to pass the Play instruction to it and get on with the rest of the program. Sometimes this can be a problem, if you want the rest of the program to continue only after the music has stopped. This is where the other format of the Play command is used.

It takes the form PLAY (x) where x is the number of the channel to be tested. For example if you want to check if channel 1 is still outputting music, use PLAY(1)

```
10 PLAY "BABABABABABABABABABABABABABABA"
20 PRINT PLAY(1)
30 GOTO 20
```

RUNning this program will display -1 on screen until the music is finished and then change to zero.

GRAPHICS COMMANDS

The COLOR command

This command is be used to change the foreground, background and border colours displayed on the screen. It takes the form COLOR a,b,c where -

a = foreground colour (used for text or drawing)
b = background colour (an area 256 x 192 pixels)
c = border colour (the rest of the screen)

Examples COLOR 1 Black foreground, background and border unchanged.
COLOR,1 Black background, foreground dndborder unchanged.
COLOR,,1 Black border, foreground and background unchanged.

First thing to remember is the American spelling of this command. If you spell it with a U then a Syntax Error will occur. The default colours (those used when the computer is first switched on) are -

	Australian SV318/328 & MSX	Japanese MSX
Foreground	15	15
Background	4	4
Border	4	7

These are also the colours used by function key F6. If you select the same colour for foreground and background, or you break a program at a point where the screen colours are unreadable, then pressing SHIFT and F6 will restore the original condition.

Note that, in a program running in the graphics screens (1 & 2 for SV and 2 & 3 for MSX), the screen will not change colour immediately following a COLOR command. The new colours will not be displayed until a SCREEN or CLS command is executed.

The PAINT command

This command is used to PAINT in a pre-defined area of the graphics screen. It takes the form -

```
PAINT(x,y),c
```

where x is the starting point across the screen and y is the starting point down from the top of the screen. The c is optional and is used to specify the colour if it is different from the current foreground colour.

PAINT(20,20) will start 20 pixels across and 20 down from the top left corner of the screen and paint in the area of the screen in the current foreground colour until it reaches the border or an area defined by the

foreground colour. For example -

This will draw a box in white and colour it in using the white foreground colour.

```
10 COLOR 15,4,4
20 SCREEN 1 : REM FOR MSX USE SCREEN 2
30 LINE(10,10)-(80,80),15,B
40 PAINT(20,20)
50 GOTO 30
```

PAINT(20,20),12 will start at the same point but will paint in the area in Dark Green until it reaches the border or an area defined in Dark Green. For example -

This will draw a dark green box on the screen and colour it in.

```
10 COLOR 15,4,4
20 SCREEN 1 : REM FOR MSX USE SCREEN 2
30 LINE(10,10)-(80,80),12,B
40 PAINT(20,20),12
50 GOTO 30
```

PAINT STEP(10,10) will start 10 pixels across and 10 down from the last referenced location. For example -

This will draw a white box on the screen and color it in.

```
10 COLOR 15,4,4
20 SCREEN 1 : REM FOR MSX USE SCREEN 2
30 DRAW"BM20,20D50R50U50L50"
40 PAINT STEP(1,1)
50 GOTO 30
```

Painting commences at the point specified and paints upwards and then down. To make this look more attractive, ensure that the starting point is either close to the top or bottom of the area to be painted. If the co-ordinates specified by the PAINT command fall on the line defining the area to be painted, then the command will appear not to have worked. This is because the paint function ceases once the outline colour is encountered.

SPRITES

PUT SPRITE continued

The co-ordinates used with the PUT SPRITE command are in relation to the top left corner of the screen. The command also has a step option

whereby the sprite will be placed at a new location relative to its last position. This takes the form -

```
PUT SPRITE 1,STEP(10,10)
```

In this example, the sprite displayed on sprite plane 1 will be moved 10 pixels (or dots) to the right and 10 down from its last displayed position.

The ON SPRITE command

This command is used to check for a Sprite collision whereby two sprites are attempting to occupy the same screen location. As mentioned before, sprites on different planes have the ability to move in front of or behind each other. Sometimes in games, such as when a missile hits the enemy, we want something to happen when these two sprites collide. This is where this command and the SPRITE ON command are used.

The ON SPRITE command has two forms - ON SPRITE GOTO and ON SPRITE GOSUB which simply mean "on one sprite colliding with another, GOTO or GOSUB to a certain section of the program". This section is normally referred to as the Sprite collision routine. The ON SPRITE command can be used at the start of the program, but before the program will GOTO or GOSUB the collision routine, the SPRITE ON command must be executed. For example -

```
10 SCREEN 1 : REM MSX USE SCREEN 2
20 ON SPRITE GOSUB 130
30 FOR I=1 TO 8 : READ A : A$=A$+CHR$(A)
40 NEXT I : SPRITE$(1)=A$
50 FOR I=1 TO 8 : READ B : B$=B$+CHR$(B)
60 NEXT I : SPRITE$(2)=B$
70 PUT SPRITE 1,(160,96),15,1
80 SPRITE ON
90 FOR I=1 TO 200
100 PUT SPRITE 2,(I,96),1,2
110 NEXT I
120 GOTO 90
130 CIRCLE (I,96),20,6
140 PAINT (I,96),6
150 RETURN
160 DATA 255,0,255,0,255,0,255,0
170 DATA 255,0,255,0,255,0,255,0
```

The ON SPRITE function is only triggered by the points of a sprite that are turned on colliding with the turned on points of another sprite. Try changing line 170 as follows -

```
170 DATA 0,255,0,255,0,255,0,255
```

RUNning the program, you will now see that one sprite passes right through the other without triggering the sprite collision routine.

QUESTIONS

1. What is wrong with this?

```
10 PRIMT "ABC"
```

2. What is wrong with this?

```
10 SCREEN 0
20 LINE(10,10)-(20,50
```

3. Will this delete lines 50 and 100?

```
DELETE -100
```

4. What will be the first new line number and increment used with this?

```
RENUM
```

5. What number will this display?

```
PRINT ASC("A")
```

(a) 72
(b) 65
(c) 81

6. What letter will this print?

```
PRINT CHR$(70)
```

(a) A
(b) Z
(c) F

7. What will this display?

```
PRINT VAL("XYZ55")
```

(a) XYZ
(b) 55
(c) 0

8. What colour will the box be painted?

```
10 COLOR 15,4,4
20 SCREEN 1 : REM MSX USE SCREEN 2
30 LINE(100,100)-(150,150),15,BF
40 PAINT(50,50),15
```

(a) Colour number 15
(b) Colour number 4

PRODUCT NEWS ...



JS-75 Wireless Joystick and Infrared Sensor

SONY REMOTE CONTROL JOYSTICK JS75

This unit consists of a handheld joystick/transmitter and a receiver unit. The receiver unit has dual cords terminating in right-angle joystick "D" connectors which plug into both joystick ports. The right-angle plugs make it difficult to connect to some computers with ports very close together but we were able to overcome this with a bit of jiggling. A permanent solution would be to remove the connectors and fit the straight type or use an extension cord.

The receiver can handle signals coming from two joysticks, and the JS-C75 joystick is available separately for this purpose. The joystick/transmitter is light and precise and weighs 170 gram including batteries. It has a range in excess of 4 metres and is effective up to 30 degrees either side of the receiver's centre line. Even turning the joystick around so that the infra-red "nose" was pointing away from the receiver did not affect performance.

Fire buttons on either side make it suitable for left or right-handed users. An indicator light is displayed on the receiver whenever each of the joysticks is used to indicate that the message has been received.

At $95 for the joystick and transmitter with $47 for the second joystick, this is not a cheap unit but for the ultimate in game playing control — it's a zip.

Review equipment supplied by Peter J. Fox Computers.

COMPUTER KIDS

Hi there, it's me again. How are things going out there across Australia? It's coming up to summer and the weather should start looking up soon.

It's time to get the shorts and T-shirts out of the drawer and get a suntan, but DO remember to wear suncream, or you'll look more like a beetroot than a kid.

Hey, does anyone remember how to swim? Summer's when you really need to know what to do in a tricky situation at the beach or the pool. It's a good idea to learn how to swim (if you don't already know) and ask Mum and Dad or your swimming teacher about some safety rules.

Most of all, though, summer is all about having Fun! So go to it! Bye for now — Wendy.

WORDSEARCH

T	R	I	H	S	T	C	M
H	F	X	D	N	A	S	A
O	P	L	S	U	N	H	E
N	M	H	U	O	Q	O	R
G	Z	A	R	T	I	R	C
S	K	T	F	U	N	T	N
B	A	T	H	E	R	S	U
G	N	I	M	M	I	W	S

Words to find -

BATHERS
FUN
HAT
SAND
SHORTS
SUN
SUNCREAM
SURF
SWIMMING
TAN
THONGS
TSHIRT

Solution next month.

If you have any computer problems or interesting things that have happened between you and a computer, write to me, and I'll put it in this column to share with fellow readers (if it's a suitable letter). The address to write to is: COMPUTERKIDS, C/o COMPUTER FORUM, P.O. Box 71, Palmyra, W.A. 6157.

HACKERS CORNER

Unfortunately our hacker, Robert Brinkworth, is no longer able to do this column for us. I think the girls and SLR5000s got to him. We do hope to bring you some more of his valuable information from time to time.

EXTRA I/O DEVICES

Although the only device supported in SV and MSX BASIC is "CAS:", there are others which are inbuilt and some can be used — for what purpose I don't know. The full list is —

LOAD "KYBD:" (SV only) —

This will allow input of a program from the keyboard. Nothing appears on the screen until zero and the ENTER keys are pressed.

SAVE	"KYBD:"	— invalid
LOAD	"CAS:"	— Fully supported
SAVE	"CAS:"	— Fully supported
LOAD	"MDM:"	— invalid (not tried this with a Modem)
SAVE	"MDM:"	— invalid (not tried this with a Modem)
LOAD	"LPT:"	— invalid
SAVE	"LPT:"	— performs the same as LLIST
LOAD	"CRT:"	— invalid
SAVE	"CRT:"	— performs the same as LIST

VIDEO RAM MEMORY MAP

The VRAM is divided into blocks according to which screen mode is active —

SV SCREEN0 & MSX SCREEN0	LOCATION	BYTES
Name table	&H0000-07FF	2048
Pattern generator table	&H0800-1000	2048
MSX SCREEN1		
Pattern generator table	&H0000-07FF	2048
Pattern name table	&H1800-1AFF	768
Sprite attribute table	&H1B00-1B7F	128
Colour table	&H2000-203F	64
Sprite generator table	&H3800-3FFF	2048
SV SCREEN1 & MSX SCREEN2		
Pattern generator table	&H0000-17FF	6144
Pattern name table	&H1800-1AFF	768
Sprite attribute table	&H1B00-1B7F	128
Colour table	&H2000-37FF	6144
Sprite generator table	&H3800-3FFF	2048
SV SCREEN2 & MSX SCREEN3		
Pattern generator table	&H0000-07FF	2048
Pattern name table	&H0800-0AFF	768
Sprite attribute table	&H1B00-1B7F	128
Sprite generator table	&H3800-3FFF	2048

MEMORY POINTERS FOR MSX / SPECTRAVIDEO

Label	MSX Addr	SV Addr	Bytes	Explanation
ERRLIN	F6B3	F7DF	2	Line in which error occured
DOT	F6B5	F7E1	2	Current line for use by LIST.
ERRTXT	F6B7	F7E3	2	Pointer used by RESUME
ONELIN	F6B9	F7E5	2	ON ERROR GOTO line
ONEFLG	F6BB	F7E7	1	Error flag=1 during error trap
TEMP2	F6BC	F7E8	2	Temporary area
OLDLIN	F6BE	F7EA	2	Old line num used by CTRL/STOP, END
OLDTXT	F6C0	F7EC	2	Old text ptr, points to next statement
VARTAB	F6C2	F7EE	2	Address of simple variables
ARYTAB	F6C4	F7F0	2	Address of dimensioned variables
STREND	F6C6	F7F2	2	End of storage pointer
DATPTR	F6C8	F7F4	2	Data pointer affected by READ
DEFTBL	F6CA	F7F6	26	Variables declaration list used by DEFSTR, DEFINT, DEFSNG, DEFDBL
SWPTMP	F7BC	F8E9	8	Value of first SWAP variable
TRCFLG	F7C4	F8F1	1	Trace flag 0=TROFF, 1=TRON
RNDX	F857	F984	8	Last random number generated

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MAXFIL	F85F	F98D	1	Highest file number
FILTAB	F860	F98E	2	Ptr to address of file data
NULBUF	F862	F992	2	Ptr to file 0 buffer
PTRFIL	F864	F997	2	Ptr to selected file data
FILNAM	F866	F99E	11	Name of file
FILNM2	F871	F9A7	11	Name of second file for NAME
NLONLY	F87C	F9B2	1	Non zero during program load
SAVEND	F87D	F9B3	2	End of memory image for BSAVE
FNKSTR	F87F	FA1E	160	Text for function keys F1 to F10
ASPECT	F931	FAC3	2	CIRCLE aspect ratio
ONGSBF	FBD8	FD73	1	ON GOSUB Flag

MSX ROM CALLS

All MSX ROM calls should be made through the BIOS entry table.

MSX1 BIOS ENTRY TABLE

0069H Name: CLRSPR
Function: Initialise sprites. Patterns=null
Sprite names=sprite plane number
Sprite colours=foreground
Vertical positions=209
Entry: SCRMOD (FCAFH)
Returns: None
Modifies: All

006CH Name: INITXT
Function: Initialise for text mode (40x24)
and set VDP
Entry: TXTNAM (F3B3H), TXTCGP (F3B7H)
Returns: None
Modifies: All

006FH Name: INIT32
Function: Initialise for text mode (32x24)
and set VDP
Entry: T32NAM(F3BDH),T32CGP(F3C1H),T32COL
(F3BFH),T32ATR(F3C3H),T32PAT(F3C5H)
Returns: None
Modifies: All

0072H Name: INIGRP
Function: Initialise for hi-res mode (Screen2)
and set VDP
Entry: GRPNAM(F3C7H),GRPCGP(F3CBH),GRPCOL
(F3C9H),GRPATR(F3CDH),GRPPAT(F3F3CFH)
Returns: None
Modifies: All

0075H Name: INIMLT
Function: Initialise for multi colour mode
Screen 3 and set VDP
Entry: MLTNAM(F3D1H),MLTCGP(F3D5H),MLTCOL
(F3D3H),MLTATR(F3D7H),MLTPAT(F3D9H)
Returns: None
Modifies: All

0078H Name: SETTXT
Function: Set VDP for text (40x24) mode
Entry: TXTNAM, TXTCGP
Returns: None
Modifies: All

007BH Name: SETT32
Function: Set VDP for text (32x24) mode
Entry: T32NAM,T32CGP,T32COL,T32ATR,T32PAT
Returns: None
Modifies: All

007EH Name: SETGRP
Function: Set VDP for hi-res mode (Screen2)
Entry: GRPNAM,GRPCGP,GRPCOL,GRPATR,GRPPAT
Returns: None
Modifies: All

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If you wish to have your user group listed in this magazine, please send details of the group and a contact name and address to The Editor, COMPUTER FORUM.

MSX users interested in forming a user group in the Adelaide area should contact
JOE ROWLEY
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We have been advised of a group in Sydney for Spectravideo owners– please contact Malcolm Samuels on 888 7664 after hours.

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PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

We have received phone calls from some people who seem to have a problem typing in some of the listings. DO NOT waste your time and ours. If the program does not work, check your typing again and, if necessary, type the line again. WE CANNOT DE-BUG PROGRAMS OVER THE PHONE.

CLOCK

HAVE THE MOST EXPENSIVE CLOCK IN THE STREET. THIS LITTLE PROGRAM WILL DISPLAY THE TIME ON THE SCREEN AND IS REASONABLY ACCURATE.

This program is suitable for use on SV318, SV328 and MSX.

```
10 SCREEN0
20 ONINTERVAL=50GOSUB80
30 INPUT"ENTER TIME H,M,S";H,M,S
40 CLS:INTERVALON
50 LOCATE0,0
60 PRINTUSING"##:##:##";H,M,S
70 GOTO50
80 S=S+1:IFS<60THENRETURN
90 S=0:M=M+1:IFM<60THENRETURN
100 M=0:H=H+1:IFH<12THENRETURN
110 H=0:RETURN
```

DENTURE ADVENTURE

USE THE JOYSTICK TO MOVE LEFT AND RIGHT AND TO MOVE TO NEXT LEVEL. THE TRIGGER CAN BE USED TO MAKE A RANDOM JUMP TO ANOTHER LEVEL.

This listing is in Spectravideo SV318/328 format. For MSX owners, see the note at the end of the listing.

```
10 REM DENTURE ADVENTURE MICHAEL PORTER
20 CLICKOFF
30 GOSUB490
40 GOSUB520
50 Q$=STRING$(43,201)
60 COLOR7:FORC=4TO19STEP5:LOCATE0,C*8:PR
INTQ$:NEXT
70 FORA=16TO248STEP16:FORB=0TO4:LOCATEA-
1,B*40+11:PRINT".":NEXT:NEXT
80 FORA=13TO239STEP239-13:FORB=12TO172ST
EP172-12:LOCATEA,B:PRINT" ":NEXT:NEXT
90 FORZ=0TO3:O(Z)=112:X(Z)=INT(RND(1)*33
):PUTSPRITE20+Z,(O(Z),Z*40+31),4,6:D(Z)=
INT(RND(1)*2):IFD(Z)=0THEND(Z)=-1ELSED(Z
)=1
100 Y(Z)=Z:NEXT:R=4:X=112:Y=4:RL=2:PLAY"
T64L1603CEFGFC":FORWQ=1TO6000:NEXT
```

```
110 STRIG(1)ON:STRIG(0)ON:STRIG(2)ON:ONS
TRIGGOSUB710,710,710
120 A=(A+1)AND1:FORZ=0TO3
130 X(Z)=X(Z)+D(Z)
140 IFX(Z)<0ORX(Z)>31THENGOSUB300
150 PUTSPRITEZ,(X(Z)*8,Y(Z)*40-1),10+Z,A
+2
160 IFY(Z)<>YTHEN200:ELSEIFX(Z)*8>X-32AN
DX(Z)*8<X+32THENMN=(MN-1)AND3:ELSE200
170 PUTSPRITE5,(X,Y*40-1),15,A
180 PUTSPRITE6,(X,Y*40-1),6,A+7
190 PLAY"O1BAG":IFMNTHEN90:ELSEPS=PS+10*
SC:GOTO40
200 IFZ=0ORZ=2THENO(Z)=(O(Z)+16)AND255EL
SEO(Z)=(O(Z)-16)AND255
210 PUTSPRITE20+Z,(O(Z),Z*40+31),4,6
220 NEXT:S=STICK(0)OR(STICK(1)ORSTICK(2)
):ONSGOSUB330,320,340,320,350,320,360,32
0
230 PUTSPRITE5,(X,Y*40-1),15,A
240 PUTSPRITE6,(X,Y*40-1),6,A+7
250 IFPOINT(X+16,Y*40+17)<>4THENSC=SC+1:
PLAY"O4L64A":IFSC=75THENSC=0:PLAY"O4GABA
BAC":LOCATEX+13,Y*40+11:COLOR4:PRINTCHR$
(201):PS=PS+750: GOTO 50
260 LOCATEX+13,Y*40+11:COLOR4:PRINTCHR$(
201)
270 IFCFTHENIFY=IYTHENIFX=112THENPLAY"O4
BAG":LS=LS+100:GOSUB370
280 IFRND(1)<.09THENGOSUB370
290 GOTO120
300 SWAPR,Y(Z):D(Z)=INT(RND(1)*2):IFD(Z)
=1THENX(Z)=0ELSED(Z)=-1:X(Z)=31
310 RETURN
320 RETURN
330 IFY=0THENRETURNELSEIFO(Y-1)<>XTHENRE
TURNELSEY=Y-1:RETURN
340 X=X+16:IFX>224THENX=224:RETURNELSERE
TURN
350 IFY=4THENRETURNELSEIFO(Y)<>XTHENRETU
RNELSEY=Y+1:RETURN
360 X=X-16:IFX<0THENX=0:RETURNELSERETURN
370 IC=(IC+1)AND3:IFIC=1THENCF=0:PUTSPRI
TE10,(0,0),4,20:PUTSPRITE11,(0,0),4,20:R
ETURN
380 IFIC=0THENCF=1:IY=INT(RND(1)*4)+1:PU
TSPRITE10,(112,40*IY+10),11,4:PUTSPRITE1
1,(112,40*IY-22),3,5
390 RETURN
400 DATA 0,0,0,0,0,0,219,108,219,108,219
,108,0,0,0,0,0,0,0,0,219,108,219,108,219
,108,0,0,0,0,0,0
```

```
410 DATA 0,0,0,0,0,0,0,0,0,0,219,108,219
,108,219,108,219,108,219,108,219,108,0,0
,0,0,0,0,0,0,0,0
420 DATA 1,128,1,128,1,128,1,128,1,128,3
,192,6,96,252,63,252,63,6,96,3,192,1,128
,1,128,1,128,1,128,1,128
430 DATA 0,0,0,0,48,12,56,28,29,184,15,2
40,6,96,12,48,12,48,6,96,15,240,29,184,5
6,28,48,12,0,0,0,0
440 DATA 63,248,31,240,31,240,15,224,15,
224,7,192,7,192,3,128,3,128,1,0,0,0,0,0,
0,0,0,0,0,0,0,0,0
450 DATA 0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,3,128,15,224,31,240,31,240,
63,248
460 DATA 255,252,255,252,255,252,255,252
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0,0,0
470 DATA 255,252,255,252,255,252,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,255,252
,255,252,255,252
480 DATA 0,0,0,0,255,252,255,252,255,252
,0,0,0,0,0,0,0,0,0,0,0,0,0,255,252,255,252
,255,252,0,0,0,0
490 SCREEN1,3:DEFINTA-Z:A=RND(-TIME)
500 FORA=0TO8:S1$="":S2$="":FORB=1TO16:R
EADN,M:S1$=S1$+CHR$(N):S2$=S2$+CHR$(M):N
EXT:SPRITE$(A)=S1$+S2$:NEXT
510 HS=0:PS=0:RETURN
520 COLOR15,4,4:CLS
530 DE$="DENTURE ADVENTURE"
540 LOCATE80,50:PRINTDE$
550 LINE(50,40)-(205,68),15,B
560 PS$="00000000":HS$=PS$
570 IFPS>HSTHENHS=PS
580 P$=STR$(PS):H$=STR$(HS)
590 PS$=RIGHT$(PS$,9-LEN(P$))+RIGHT$(P$,
LEN(P$)-1)
600 HS$=RIGHT$(HS$,9-LEN(H$))+RIGHT$(H$,
LEN(H$)-1)
610 LOCATE71,80:PRINT"HIGH SCORE-"HS$
620 LOCATE71,90:PRINT"LAST SCORE-"PS$
630 ST$="PRESS 'SPACE' TO START"
640 LOCATE63,150:PRINTST$
650 IFSTRIG(0)THENCLS:PS=0:SC=0:MN=3:RET
URN
660 B=(B+1)AND7:IFB=0THENA=(A+1)AND1
670 PUTSPRITE5,(128-16,15*40+19),15,A
680 PUTSPRITE6,(128-16,15*40+19),6,A+7
690 GOTO650
700 GOTO700
710 IFRLTHENRL=(RL-1)AND1:Y=INT(RND(1)*5
):X=INT(RND(1)*15)*16:RETURN
720 RETURN
```

NOTE: FOR MSX CHANGE THE FOLLOWING LINES - 20, 50, 60, 70, 80, 250, 260, 490, 520, 540, 610, 620, 640 and 650

```
20 OPEN"GRP:"AS#1
50 Q$=STRING$(32,219)
60 COLOR7:FORC=4TO19STEP5:PRESET(0,C*8):
PRINT#1,Q$:NEXT
70 FORA=16TO248STEP16:FORB=0TO4:PRESET(A
-1,B*40+11):PRINT#1,".":NEXT:NEXT
80 FORA=13TO239STEP239-13:FORB=12TO172ST
EP172-12:PRESET(A,B):PRINT#1," ":NEXT:NE
XT
250 IFPOINT(X+16,Y*40+17)<>4THENSC=SC+1:
PLAY"O4L64A":IFSC=75THENSC=0:PLAY"O4GABA
BAC":PRESET(X+13,Y*40+11):COLOR4:PRINT#1
,CHR$(219):PS=PS+750: GOTO 50
260 PRESET(X+13,Y*40+11):COLOR4:PRINT#1,
CHR$(219)
490 SCREEN2,3:DEFINTA-Z:A=RND(-TIME)
520 COLOR15,4,4:SCREEN0:SCREEN2
540 PRESET(60,50):PRINT#1,DE$
610 PRESET(71,80):PRINT#1,"HIGH SCORE-"H
S$
620 PRESET(71,90):PRINT#1,"LAST SCORE-"P
S$
640 PRESET(63,150):PRINT#1,ST$
650 IFSTRIG(0)THENSCREEN0:SCREEN2:PS=0:S
C=0:MN=3:RETURN
```

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